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Final Report

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Introduction

This overview details the structured application of the Design Thinking methodology to create a tailored solution addressing specific user needs. The report aims to thoroughly address the phases of two foundational pillars of project design, *Desirability* and *Feasibility,* integral in ensuring that solutions are user-centered and implementable within realistic constraints.

Desirability represents the human aspect of innovation, focusing on understanding potential users and identifying their needs by answering questions such as:

- **What resonates most with users?**
- **Are there unaddressed needs and desires?**
- **How could these needs be met effectively?**
- **How important are these needs to our target users?**
- **What evidence supports these insights?**

Innovators aim to identify who their users are, what they desire, value, and what holds significance for them. Various methods were outlined to explore the emotional and psychological aspects of users' needs, first to identify and locate users and their needs, then to develop hypotheses around them, test them, craft need statements, and evaluate them iteratively. The process begins with generating various options and concludes by narrowing them down and shifting from divergent to convergent thinking.

Feasibility, on the other hand, emphasizes the technical and operational aspects, determining what is realistically achievable to meet the identified needs while still aligning with users' emotions and values. This section explores methods for generating, describing, and evaluating solutions, along with presenting the corresponding outcomes. This way, the most suitable and implementable solutions to user needs are effectively identified, analyzed, and assessed.

Given the constraint of designing solutions within the broad context of the arts, making a well-informed and grounded choice for the project was essential. We chose to focus on the **music industry** as it is a vibrant and influential sector within the arts, valued at $26 billion in the global market. Music has the power to connect people across cultures, evoke emotions, and create shared experiences, making it a uniquely impactful and universal form of expression.

1. Desirability

The first step into the desirability phase is to showcase the detailed approaches and techniques used to explore the music industry, aiming to identify the users and their key necessities. The process is precisely organized as below.

- **First**, we select and describe the methods and tools that help us identify potential users and their possible needs.
- **Next**, we analyze the found users and needs, exploring them using appropriate techniques to uncover meaningful insights.

- **Finally**, results are presented, summarizing the key outcomes and insights gained to assess desirability.

1. 1. Methods to Identify Users and Needs

In this section, we highlight the approaches used to identify users and understand their needs. We have chosen to apply and iterate various techniques to support effective decision-making in recognizing users and assessing their potential needs.

Brainstorming

Brainstorming is the first approach applied. This creative and collaborative technique allowed us to explore a wide range of ideas in an open and dynamic environment. Brainstorming was particularly effective for generating diverse perspectives on the music industry, as it encourages free-flowing contributions from all participants, fostering innovation and inclusivity.

To guide our session, we established some rules:

1. Avoid dismissing or critiquing each other's suggestions during the idea generation session to maintain a positive and open atmosphere.
2. Actively listen to one another and have an in-depth understanding of different perspectives and initial suggestions.
3. Focus on *quantity over quality* at first.
4. Refrain from judgment until the session's end to allow ideas to flow freely.

The ideas generated during the brainstorming sessions were recorded and mapped using a digital whiteboard. Users were grouped into clusters that reflected similar themes or user groups. At the end of each session, team members were invited to share their thoughts on how the brainstorming process went. Once all proposals were consolidated, the team collectively evaluated them by voting. Any ideas that did not gain the support of the majority were set aside.

After a couple of rounds of brainstorming, we ended up with four concrete examples of users in the music field:

- **Vinyl collectors**: select rare and limited-edition vinyl records to expand their collections.
- **Songwriters**: collaborate with other artists and producers to get their songs recorded and released.
- **Emerging musicians**: play original songs during gigs to build their fan base and gain recognition.
- **Hidden-gem seekers**: explore different genres of music, seeking out and championing new and lesser-known music.

Role-storming

During role-storming sessions, participants take on specific roles to generate ideas and explore challenges from different perspectives. By stepping into the shoes of distinct personalities of the music environment, participants can approach problems creatively and uncover insights that might otherwise be overlooked. This technique helped us to broaden our thinking by encouraging us to break away from our usual thought patterns.

The team began by adopting the four user categories identified during brainstorming as a starting point. These categories were treated as distinct user roles with different perspectives, for which we explored potential worst-case scenarios. Here is an example of how the role-storming sessions were conducted.

- As a **vinyl collector**, I have trouble finding unique pieces to add to my personal collection.
- I am a **songwriter**, and I could not get my songs recorded and released. Nobody wants to collaborate with me.
- As an **emerging musician**, even if I play gigs, I fail to get an audience and be recognized on the streets.
- As a **hidden gem seeker**, I love discovering unknown music, but sometimes I do not know how.

To ensure the robustness of this approach, we conducted multiple sessions focusing on users excluded in earlier brainstorming rounds, evaluating alternative perspectives and solutions that eventually revealed themselves as irrelevant to our scope. Role-storming allowed us to identify and examine problems in the music industry through various simple yet meaningful lenses, deepening our understanding of potential users and needs in the unique contexts in which they operate.

5 Whys Method

Root-cause analysis is a systematic method used to identify the underlying causes of a problem rather than just addressing its symptoms. It seeks to uncover what led to an issue by examining causality and asking iterative questions. Among the various techniques within RCA, the *5 Whys method*[1] is particularly effective. It involves asking "why" repeatedly—typically five times or more—to drill down to the core of a problem. This method is simple yet powerful, as it helped the team explore and uncover root causes that might not have been immediately apparent.

The 5 Whys approach is widely applied across industries for troubleshooting and problem-solving. Its effectiveness lies in its simplicity and flexibility, making it suitable for exploring user-centric challenges like those in our project. Below is an example of how we implemented the method in our case study based on the assumptions made through role-storming sessions.

Problem: "*Why do songwriters struggle to collaborate with other artists?*"

1. **Why?** They lack access to a network of like-minded collaborators.
2. **Why?** There are limited spaces specifically designed for connecting songwriters and artists.

[1] https://businessmap.io/lean-management/improvement/5-whys-analysis-tool

3. **Why?** Most existing labels prioritize finished music and established creators rather than the creative process.
4. **Why?** The industry focuses more on showcasing results rather than facilitating the collaborative journey.
5. **Why?** There's a gap in tools that cater to creators in the early stages of development and collaboration.

While the 5 Whys technique is a valuable tool for uncovering root causes, it is not without its limitations. Issues such as stopping at surface-level symptoms, reliance on the problem solver's existing knowledge, and variability in results highlight the need for careful application.

To mitigate these limits, we validated each response by discussing it as a team and cross-referencing it with any available evidence or prior insights. This collaborative validation ensured a more accurate and reliable analysis, allowing us to delve deeper into the underlying causes and maintain consistency throughout the process.

Web Search

Web search is fundamental to research and gathering diverse information from online sources, providing a foundation for exploring ideas, trends, and potential focus areas. It offers access to many resources, including industry reports, academic publications, articles, and blogs, enabling researchers to uncover valuable insights and contextual knowledge.

In this project, web research served as a tool to broaden our knowledge and again validate our insights by exploring the music industry through the perspective of the *World Wide Web*. Our approach included:

- **Exploring industry trends**: we examined articles, reports, and online discussions to understand key developments and challenges within the music field.
- **Investigating existing tools and platforms**: research was conducted to identify current technologies used in the industry, highlighting gaps and opportunities for innovation.
- **Learning from case studies**: successful initiatives and projects were reviewed to gain insights into effective strategies and approaches within similar contexts.
- **Strategic search**: we used critical thinking, strategic searching, and source evaluation to ensure credibility and relevance. Broad queries on *Google* were drilled down to specific ones and then critically evaluated based on *objectivity*, *currency*, and *source relevance*.

This approach provided a structured framework for gathering and synthesizing new information. Through this process, a new user category emerged: **independent label owners**. The research highlighted several key challenges they face. The industry's fragmentation and heavy reliance on streaming platforms make it difficult for these labels to connect with specific audiences and achieve sustained growth. Additionally, monetizing streaming royalties often falls short of adequately supporting emerging artists. Compounding these issues, the rise of AI-generated music and fraud

requires independent labels to adapt swiftly to a rapidly evolving and increasingly competitive landscape.

Interviews

Lucky chances to converse online (via *Instagram* chats) with various figures active within the music industry, such as musicians, label representatives, and fans, brought us to conduct interviews, which allowed us to capture diverse insights. Through casual questions, we were able to get information about their motivations, frustrations, and expectations around the process of making and finding music. These insights informed the development of empathy maps, helping us better understand how different users approach the music creation and discovery. A recurring theme emerged: *musicians new to the scene often face significant challenges in finding and connecting with audiences*. These struggles reflect broader systemic issues, such as the overwhelming volume of music available and the limited opportunities for emerging artists to gain visibility.

1. 2. Methods to evaluate Users and Needs

The *evaluation phase* is a critical step in ensuring the user-need relationships uncovered in earlier stages are validated. By grounding the process in evidence-based testing, this phase ensures that the needs identified truly align with user realities. It begins with formulating *need hypotheses*—assumptions about these user-need connections—which are then put to the test through carefully designed methods presented here. The insights gained from this process confirm or challenge these hypotheses, providing a solid foundation for further development in the *feasibility stage*.

Need Hypotheses

Need hypotheses were formulated in the most basic form to provide a starting point for the team. Based on the insights collected through the identification phase, we generated simple "need evidences" to use as starting hypotheses, to confute or confirm during the evaluation process. Need hypotheses are to be found in *Table 1* of the result section.

SWOT Analysis with Chat-GPT

SWOT analysis is a strategic planning tool used to assess the **strengths**, **weaknesses**, **opportunities**, and **threats** associated with a specific goal, user group, or situation. It provides a comprehensive overview of internal and external factors that may influence the subject of the analysis, enabling informed decision-making. The four components explained are shown in the figure.



Figure 1 – How to do a SWOT Analysis

- **Strengths**: internal advantages or resources that can help achieve objectives.
- **Weaknesses**: internal challenges or limitations that may hinder progress.
- **Opportunities**: external factors or circumstances that can be leveraged for benefit.
- **Threats**: external risks that could negatively impact outcomes.

In our case study, the analysis was performed manually through group discussions. Each member contributed insights for each user group, focusing on the unique attributes, challenges, opportunities, and risks associated with their needs. These discussions highlighted the factors most relevant to our project and provided a foundation for a structured analysis.

To refine our findings, we utilized **generative AI** to ensure a thorough and unbiased evaluation. Using the following prompt in ChatGPT, we analyzed the potential users:

*"We have identified five potential user groups within the music industry: **Vinyl collectors** (need access to rare and limited-edition records), **Songwriters** (need collaboration opportunities with artists and producers), **Emerging musicians** (need to connect with a wider audience), **Hidden-gem seekers** (need to discover new and lesser-known music), and **Independent label owner** (need better audience targeting tools). Can you perform a SWOT analysis for these user groups, highlighting which have more strengths, weaknesses, opportunities, and threats? Assign scores to determine which user group may be the most advantageous for a solution tailored to the music industry. Consider internal factors (e.g., user passion, resources) and external factors (e.g., market trends, competition)."*

Maslow's Pyramid

Maslow's pyramid is a framework that categorizes human needs into five hierarchical levels. Starting from the base, the pyramid includes physiological needs, followed by safety needs, belongingness and love needs, esteem needs, and, at the apex, self-actualization needs (*figure 2*). The lower-level needs are foundational and must typically be fulfilled before addressing higher-level, more abstract needs. This theory is widely used in various fields, including psychology, marketing, and design, to understand better and prioritize user requirements.



Figure 2 – Maslow's Hierarchy of Needs

For this project, we matched the needs of identified user groups from our Role Storming session with the levels of Maslow's Pyramid. This helped us determine the alignment of these needs with Maslow's hierarchy, offering a structured way to evaluate their importance. To perform this analysis, we combined personal insights with input from generative AI. In particular, we used the following prompt:

Interpret the role of a Design and Psychology expert. Your task is to evaluate needs related to specific users using Maslow's pyramid. The pyramid has five levels: ***physiological needs*** *(score 5),* ***safety needs*** *(score 4),* ***belongingness needs*** *(score 3),* ***esteem needs*** *(score 2), and* ***self-actualization needs*** *(score 1). Evaluate each user and their connected need, assign it to a level, and explain your reasoning. Prepare the output in a table format for use in a report.*

The decision to assign scores to user needs was made collaboratively, grounded in our understanding of Maslow's hierarchy. The five-level version of the pyramid was selected to ensure clarity and alignment with the filtered user groups identified in previous methods. Using ChatGPT helped validate our assessments.

> *Vinyl collectors* align with **esteem needs**, as their pursuit of rare records reflects a desire for recognition and accomplishment.
>
> *Songwriters* fall under **self-actualization needs**, as their focus on creative collaboration supports personal growth and artistic expression.
>
> *Emerging musicians* primarily address **belonging needs** by building an audience, which fosters emotional connections while indirectly supporting financial stability—*physiological need*.
>
> *Independent label owners*, on the other hand, primarily deal with **esteem needs**, as their focus on discovering and promoting unsigned talent stems from a desire for recognition within the industry and the achievement of building a successful, respected label.

Finally, *hidden-gem seekers* also align with **belonging needs**, as their passion for discovering new music creates a sense of connection and shared identity within a broader community.

Empathy Map

The empathy map is a user-centered design tool that helps teams better understand their users by visualizing their experiences, behaviors, and emotions. It is divided into four quadrants—**Say**, **Think**, **Do**, and **Feel**—to create a holistic view of the user's perspective. This method is particularly valuable for identifying pain points, motivations, and opportunities for creating user-focused solutions. By capturing insights in this structured way, teams can align their efforts to address user needs effectively.

For this project, we used empathy mapping to delve deeper into the perspectives of our identified user groups within the music industry. The insights were gathered from role-storming exercises, brainstorming, and surveys, and the results are summarized in *Table 2* of the result section.

Surveys

Surveys are a versatile research tool used to collect structured information about specific phenomena, ensuring that the data obtained is comparable and actionable. For this project, we designed a concise questionnaire aimed at individuals in the music industry, including musicians, songwriters, and music fans. The questions were crafted to be simple and clear, allowing participants to complete the survey quickly and accurately.

The questionnaire consisted of **10 questions**, including multiple-choice and open-ended formats. The first section sought *demographic information*, such as age and background, to establish a profile of the respondents. The next set of questions focused on identifying *individuals' key needs and challenges*. Multiple-choice questions asked respondents to rank common challenges such as *financial constraints*, visibility status, and *possibilities for accessing opportunities*. Open-ended questions encouraged participants to elaborate on their **unique perspectives**, sharing their most pressing needs or aspirations.

The survey received **120 responses** from participants aged 18 to 50, with the majority falling between 20 and 30. This diverse age range provided a broad spectrum of insights, reflecting different stages of engagement with the music industry. The survey was distributed through online platforms, music-related communities, and social media to ensure a varied respondent pool. This approach aimed to capture whether the identified needs represented individual challenges or broader trends shaped by cultural and societal factors.

Need Hypotheses

Drawing from insights gained through SWOT analysis, Maslow’s pyramid, and survey data, the previously stated need hypotheses were refined to better capture the user needs and desires. For example, vinyl collectors’ hypothesis to seek rare records was refined to emphasize recognition and status through the collection, hence clarified to address both the **emotional needs** and **practical goals**. The refining process involved each user group, allowing us to ensure that the conclusions

were consistent, accurate and aligned with the data. Refined need hypotheses are to be found in *Table 3* of the result section.

1. 3. Need Statements

After validating the initial need hypotheses, the next step is to refine these hypotheses into clear and actionable **need statements**. This process involves taking the validated insights into the core needs of each user group and transforming them into accurate descriptions that clearly state the users' needs, precising each into three categories: **user**, **need**, **goal**. The need statements provide a solid foundation for developing targeted solutions in the next phase of the project, *feasibility*, ensuring that the proposed solutions address real user challenges and align with their deeper motivations. Need statements are presented in *Table 4* of the result section below.

1. 4. Results Section

Users and Needs

Users and needs are identified by adopting the method shown in Section 1.1. After the first phase, ideas are generated, filtered, and shaped according to the scope of our project. In the figure below, there is the table that summarises the output of the identification process:

User	Need
Vinyl collectors	Need access to rare and limited-edition vinyl records to expand their collections.
Songwriters	Need opportunities to collaborate with artists and producers to get their songs recorded and released.
Emerging musicians	Need to reach and connect with a wider audience to build their fan base and gain recognition.
Independent label owners	Need effective ways to scout unsigned talent and promote their artists in a competitive market.
Hidden-gem seekers	Need to discover new and lesser-known music to expand their listening experience.

Table 1 - Users and Needs

Empathy Map

User Group	Say	Think	Do	Feel
Vinyl collectors	"I'm always on the hunt for exclusive releases.", "Limited-edition vinyl is what keeps my collection unique."	"Will I ever find that one rare pressing?" "Are these prices worth it for my collection?"	Attend vinyl fairs, browse online marketplaces, and network with other collectors.	Excited when discovering new pieces, frustrated when records are unavailable and proud of their collection.

Songwriters	"I just need the right artist to hear my work." "Collaborating makes my music come alive."	"How can I make my songs stand out?" "Where can I find artists who share my vision?"	Share demos online, attend songwriting workshops, and network with musicians.	Inspired by creativity, but often isolated or overlooked without collaboration opportunities.
Emerging musicians	"I want my music to reach more people." "Getting noticed is the hardest part."	"How do I stand out in a crowded space?" "Am I doing enough to grow my fan base?"	Perform live, share music on streaming platforms, and engage with fans on social media.	Excited by gaining new followers, discouraged by slow progress or lack of visibility.
Independent label owners	"Finding the right talent is everything." "Promoting our artists is an uphill battle."	"What makes this artist unique?" "How can I stretch my budget to promote effectively?"	Attend local shows, browse platforms for talent, plan targeted marketing campaigns.	Driven to discover the next big act, pressured by limited resources and competition.
Hidden-gem seeker	"I love finding hidden gems." "Mainstream music doesn't excite me anymore."	"Where can I find unique, authentic artists?" "What if I miss out on great music?"	Explore curated playlists, follow independent artists, and attend intimate live performances.	Thrilled by discovering new music, overwhelmed by the volume of options.

Table 2 - Empathy Map

Need Hypotheses

User Group	We believe that the user...	So, if we...	Then we will see...
Vinyl collectors	value access to rare and limited-edition vinyl records to expand their collections.	provide clear and reliable information about the availability of such records	increased engagement, with collectors showing positive feedback and interest in these offerings.
Songwriters	need opportunities to collaborate with artists and producers to get their songs recorded.	improve their visibility and create opportunities to connect with potential collaborators,	increased collaboration activity and recognition for songwriters within the music industry.
Emerging musicians	need to reach and connect with a wider audience to build their fan base and gain recognition.	provide opportunities for them to showcase their music to new audiences,	growth in their fan base and an increase in their visibility and audience engagement.

Independent label owners	need effective ways to scout unsigned talent and promote their artists in a competitive market.	make it easier to discover unsigned talent and provide tools for artist promotion,	active engagement from labels scouting talent and effectively promoting their roster.
Hidden-gem seekers	want to discover new and lesser-known music to expand their listening experience.	make discovering unique and lesser-known artists easier,	increased engagement from fans, reflected in listening activity, playlist creation, and positive feedback.

Table 3 - Need Hypotheses

Need Statements

The collected methods and results made us finally refine our understanding of our user base and their needs. This section presents an initial analysis of emerging musician personas and their primary needs, based on survey data and Maslow's Hierarchy of Needs.

User	Need	Goal
Vinyl collectors	Access to rare and limited-edition vinyl records	To expand their collections and feel pride in owning unique and valuable items.
Songwriters	Opportunities to collaborate with artists and producers	To get their songs recorded and released, enabling creative expression and professional growth.
Emerging musicians	Reach and connect with a wider audience	To build a loyal fan base, gain recognition, and achieve career stability in the music industry.
Independent label owners	Effective ways to scout unsigned talent and promote their artists	To discover promising new talent and enhance the visibility and success of their roster.
Hidden-gem seekers	Discover new and lesser-known music	To expand their listening experience and feel connected to unique and authentic music communities.

Table 4 - Need Statements

2. Assessment



Figure 4 – Parameters to prioritise need statements

This final stage aims to weigh and rank each identified need, allowing us to uncover the top priority for every user group. We used three main variables for evaluation:

A. Number of Users

This variable helps gauge the potential coverage of each need. The estimation of user numbers for this project was based on data from industry reports, ññcredible platforms, and reasonable assumptions. Where specific data was unavailable, we estimated user numbers based on music industry reports, surveys, and general statistics on emerging artists.

Vinyl collectors are estimated at approximately 500,000 globally, derived from reports on the resurgence of vinyl sales and the size of niche collector communities.[2][3] For **songwriters**, platforms like ASCAP and BMI collectively represent around 1 million active members, making this a reasonable estimate for active songwriters globally.[4] **Emerging musicians** were estimated using data from Chartmetric, which tracks 1.3 million artists, with 87.6% classified as "undiscovered", resulting in approximately 1.14 million emerging musicians worldwide.[5] For **independent label owners**, data from associations such as the Association of Independent Music (AIM) and Merlin Network support an estimate of around 50,000 globally, encompassing both active and small-scale labels.[6] Finally, for **hidden-gem seekers**, given the vast number of global music consumers, particularly on streaming platforms like Spotify, hundreds of millions discover new and lesser-known music, but only a small portion of those people are truly dedicated to finding unpopular artists on a daily basis. These estimates provide a foundational understanding of the user groups' scale and diversity.

B. Importance

Each factor is scored on a scale of **1 to 10**, and the total importance is calculated as the average of these three factors:

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[2] https://www.discogs.com/digs/collecting/750-million-records-in-global-collection
[3] https://www.statista.com/chart/2967/worldwide-vinyl-sales
[4] https://bigtimemusicians.com/how-many-official-songs-are-there-in-the-world/
[5] https://www.hypebot.com/hypebot/2024/02/87-6-of-all-musicians-releasing-music-are-undiscovered.html#:~:text=About%2012%25%20of%20developing%20artists,or%20Superstar%20(top%201%2C500)
[6] https://www.musiciansunion.org.uk/

1. **Impact on users' lives**:
    - How much they need, if met, would improve the user's life or situation.
    - Scored based on qualitative analysis (e.g., high for financial or audience growth needs).
2. **Alignment with Maslow's hierarchy**:
    - Needs aligned with higher-priority levels (e.g., physiological or belonging) are scored higher
3. **Benefits in addressing major challenges**:
    - How effectively the need addresses significant barriers for the user (e.g., lack of resources, visibility).
    - Scored based on how directly solving the need resolves the user's major challenges

Each ranking captures how pressing and impactful each need is from the artist's viewpoint:

User	Impact on user's life	Maslow alignment	Addressing challenges	Total
Vinyl collectors	5	6	4	5.0
Songwriters	7	4	6	5.7
Emerging musicians	9	8	9	8.7
Independent label Owners	6	6	7	6.3
Hidden-gem seekers	5	7	8	6.6

C. Confidence

Confidence scores represent the reliability of the data used to evaluate each need. These scores are crucial for determining how much trust can be placed in the conclusions drawn from the analysis. The confidence metric is based on three main factors:

- **Reliability of data source**: The credibility of the data source, with higher scores given to well-established, verified sources like national reports, industry databases, and large-scale surveys.
- **Sample size representativeness**: The diversity and size of the sample used to collect data, with higher scores for large and diverse samples that reflect the user group accurately.
- **Alignment with multiple sources**: The degree to which the data is supported by multiple independent sources, ensuring consistency and reducing bias.
- **Team evaluation**: The team's collective agreement on the reliability and relevance of the data for each user group, scored after group discussions.

Each factor is scored on a scale from 0 to 1, and the overall confidence score is calculated as the average of these four factors. Higher scores indicate a greater degree of reliability and

trustworthiness of the data, ensuring a robust basis for prioritizing user needs. This approach helps mitigate risks associated with unreliable or incomplete data, providing a strong foundation for the final analysis.

User	Reliability of Data Source	Sample Size Representativeness	Alignment with Multiple Sources	Team evaluation	Total
Vinyl collectors	0.7	0.6	0.7	0.6	0.65
Songwriters	0.8	0.7	0.8	0.7	0.75
Emerging musicians	1.0	0.9	1.0	0.9	0.95
Independent label owners	0.9	0.8	0.9	0.5	0.77
Hidden-gem seekers	0.5	0.7	0.5	0.8	0.62

Number of users x Importance x Confidence

User	Number of users	Importance	Confidence	Score
Hidden-gem seekers	2 000 000	6.6	0.62	8 184 000
Emerging musicians	1 140 000	8.7	0.95	9 422 100
Songwriters	1 000 000	5.7	0.75	4 275 000
Vinyl collectors	500 000	5.0	0.65	1 625 000
Independent label owners	50 000	6.3	0.77	242 550

3. Final Decision

Assessment results led the team to the final decision. The **key user** we'll focus on is *emerging musicians*, with the **key need** of *increasing visibility* and *exposure*. This represents the belonging basilar need that would lead this category of users to create a loyal fan-base and accelerate their careers. However, this comes with challenges, including the evolving and saturated music market, where established artists dominate, making it hard for newcomers to gain attention. Plus, the

changing habits of the audience, demanding some more in the digital music market. We also need to create a scalable solution that adapts to *different music genres* as well as artist types, whether local performers or niche artists. Addressing these challenges will require creative, user-centered solutions, based on our new deep understanding of emerging artists' requirements.

4. Feasibility

The feasibility phase takes the users' need statement defined in the desirability section as a starting point. Our project focuses on leveraging technological and process-based solutions for the critical challenge of *gaining visibility in the competitive music industry endured by emerging musicians*. To an initial burst of ideation using different methodologies, it followed a detailed evaluation, allowing us to identify the most promising final solutions.

4. 1 Methodology Overview

Various methods were used to first generate and identify potential technologies and their process, and second to evaluate them as feasible solutions. The aim was to ensure a comprehensive approach and bias reductions. You can find below:

- **A technological solution generation section**: about identifying and exploring potential technologies to address musicians' needs.
- **A process solution identification phase** of defining workflows for implementing the chosen technologies effectively.
- **Experimentation** and testing of proposed solutions through iterative trials.

This structured approach ensures all options are evaluated, practical processes are developed, and solutions are aligned with the core need of enhancing visibility for emerging musicians.

4. 2 Technological Solution Generation

This phase is a crucial initial step in the feasibility process, where the focus shifts from understanding user needs to exploring a wide array of technological possibilities. The objective is to begin with a divergent phase, casting a broad net to identify as many potential solutions as possible. By utilizing both qualitative and quantitative methods, this phase seeks to identify and prioritize technologies that hold the greatest promise for addressing the challenges at hand. It follows an overview of methods involved.

Trust Workshops Challenges (with AI)

During the lectures, the team conducted a series of workshops, in order to evaluate trust in AI systems while exploring potential solutions to problems assessed on our need statements. The first workshop session aimed to generate as many possible solutions as we could by providing AI with a broad, open-ended prompt. In the second phase, we refined the results by applying three evaluation filters: *technical feasibility*, *product uniqueness*, and *fitting with the need*. These workshops served as tools to leverage AI for developing effective, user-centered solutions.

Starting from general ideas to be found here below, the technologies solution identification section continues with other methods to identify the solutions most practical, user-friendly, and aligned with emerging musicians' needs.

Possible solutions	Inputs	Process	Outputs
Gamified participation	Audience engagement tools	Develop a platform offering rewards (e.g., badges, content)	Increased fan engagement and loyalty
Gig-matching platform	Venue and musician profiles	Match musicians with venues by genre, location, and size	More performance opportunities
Feedback tool for listeners	Music tracks and rating criteria	Allow listeners to rate based on lyrics, melody, etc.	Actionable feedback for musicians

Role-Storming Sessions

The team engaged in role-play once again, to focus on finding solutions to the needs already identified during the desirability part. This approach sparked creative thinking and helped pinpoint the specific constraints faced by emerging musicians. However, the ideas generated were still mostly centered on general strategies rather than technological solutions.

Brainstorming

Collaborative brainstorming sessions involving 10 emerging musicians, ChatGPT, and computer science students generated over **20 innovative ideas**. The feasibility of these ideas was assessed based on technical and economic constraints. Some ideas were relatively easy, such as T-shirts with the musicians' QR codes; others were interesting DIY ones, like treasure hunts, as a way to help listeners discover physical gigs. Simpler ones alternated with more complex ones, all yielding a wealth of possibilities which were then organized into categories to streamline the selection process.

Google Trends Analysis

This time, we explored the web through *Google Trends* following a fresh intuition, aiming to determine whether the listening audience might be shifting away from in-door gigs in favour of live streaming music and online performances. Google Trends data for *"Live Music Streaming"*, *"Interactive Concerts"* and *"Music Livestream"* revealed significant insights into a distinct demand for digital music experiences:

- **Strong demand for live music streaming**: "Live Music Streaming" consistently shows high public interest, with regular spikes, indicating stable demand for accessible live music online.

- **Emerging interest in interactive experiences**: Increasing searches for "Interactive Concerts" and "Music Livestream" suggest a growing interest in immersive, interactive music experiences, reflecting shifting digital preferences.

Google Trends underscored the evolving nature of the music market. From a feasibility standpoint, these findings could only serve as a reminder to prioritize decisions that resonate with emerging musicians' audience's needs and expectations.

"Predicting Artists' Success" Dataset Analysis and Data Understanding

To uncover insights into trends and already-existing tools relevant to emerging musicians, we conducted dataset analyses on **musicians success' predictability** from streaming platforms, in this case: *Spotify*. The goal was to identify other key factors influencing the growth in popularity of artists, to adopt a more data-driven approach to our solution-finding processes. The dataset analysis' first step was to define "emerging musicians" based on data about the *popularity score* and the *number of followers*. Our findings revealed that a musician's popularity score significantly impacted both audience engagement and the number of followers, confirming the need to focus on this category. Additionally, we analyzed which genres were most popular in terms of *trends* and *follower counts*. It became clear that musicians in widely recognized genres, such as pop, tended to have larger followings. Genre-specific success metrics could add value to our potential solution.

For example:

```
As we can see from the popularity histogram (figure 6 of the result section), the distribution is gaussian centered at 50 plus a constant shift for the low popularity with a peak at 0 popularity. That shows that low popularity artists represent a big portion of the artists (the 20 percentile of the data) present on Spotify.
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While if we look at the chart for the genre of the artists with low popularity score(figure 7 of the result section) we can see an almost constant distribution among the genres, with few exceptions with a particular high count, suggesting a particular unpopularity for such genres (like avant-garde, contemporary classical,early modern classical...) showing people preferences.
```

The data analysis confirms that there is an audience interested in unrepresented musicians, seeing the gaps in exposure resulting from data analysis. Low popularity among a substantial segment of artists, the lack of correlation with activity levels, and the challenges faced by niche genres all emphasize the need for environments that enhance visibility for artists and foster connections. Establishing such an opportunity could provide underrepresented artists the spaces to grow, collaborate, and reach their audiences effectively.

Fishbone Diagram

A *fishbone diagram*[7] served to better structure the thought flows, and to identify the root causes of low visibility for emerging musicians. The most promising insights were selected and categorized

[7] What is a Fishbone Diagram? Ishikawa Cause & Effect Diagram | ASQ

into three areas, fan engagement, performance opportunities, and feedback mechanisms, to better define the solutions previously brainstormed into clearer, logical ones. The fishbone diagram revealed key factors for musicians' low visibility:

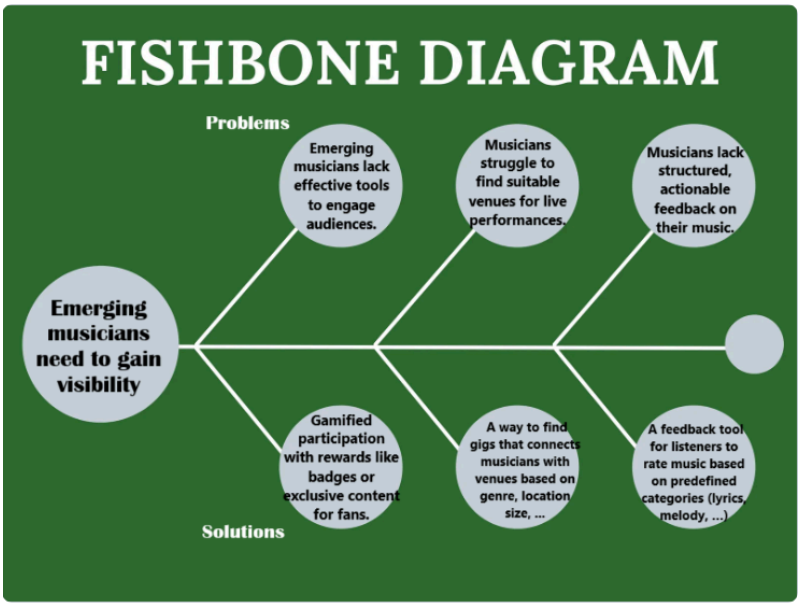


Figure 3 – Fishbone Diagram for problems and solutions relative to emerging musicians

From this diagram, it emerges that potential solutions include creating fun, gamified ways to connect with fans, developing platforms that match musicians with the right venues, and offering tools for listeners to provide constructive feedback. By focusing on these areas, we can help emerging artists build stronger connections with their audience, access better performance opportunities, and grow through actionable insights on their work.

How-Now-Wow-Huh?-Ciao Matrix

Another helpful method to categorize the potential solutions was the *How-Now-Wow-Huh?-Ciao Matrix*[8]. This was based on ideas' feasibility, uniqueness, and value to the user experience. Here's an example of how we placed different ideas within the matrix:

- **"How"**: Innovative ideas that are still feasible.
- **"Now"**: Simple and commonly used ideas that are easy to implement.
- **"Wow"**: Easy-to-implement ideas that enhance the user experience.
- **"Huh?"**: Ideas too similar to others already developed.
- **"Ciao"**: Ideas that are both uninteresting and difficult to implement.

This approach provided a more organized overview of all potential solutions based on all the insights from previous solutions findings, allowing us to immediately focus on the "Wow" ideas, while eliminating those in the "Ciao" category due to their limited impact and implementation challenges.

Results

[8] How|Now|Wow (Ciao!). Design is a team sport. Collaboration... | by Joelle Hagen | Medium

At the end of the solution identification process, we arrived at **three top alternatives** tailored to the needs of emerging musicians still remaining cost-effective, simple to implement, and maximizing audience engagement. After analyzing various options, we selected the following three solutions: a **music treasure hunt**, an **online musical advent calendar**, and **postcards for concerts**. Each option leverages creative and accessible methods to help emerging musicians build visibility and foster meaningful relationships with their audience.

4. 3 Process Solution Identification

The **process solution identification** phase builds upon the technologies identified earlier by detailing how they will be implemented. The three optimal solutions previously chosen need to be validated through this phase as the most practical, user-friendly, and aligned with the needs of emerging musicians.

Flowcharts

Flowcharts were instrumental in visualizing the step-by-step technological implementation of our solutions. We began by assessing the different phases of how each solution would work, breaking down the entire process into smaller, manageable steps. For instance, the **advent calendar** concept was divided into several phases: a "*preparation phase*", where we recruited musicians, collected their music, and obtained permissions for content sharing; the "*product design phase*", where we selected the format (*website*) to use, and organized the content and schedule. Next, the "*promotion phase*" would kick off, followed by the "*launch phase*", where daily content would be shared on the website, and audience feedback would be monitored. By using flowcharts, we were able to map out each task in the process, connecting them in a logical order, ensuring clear visualization of how each solution would be implemented. Below is the flowchart relative to the product design phase.

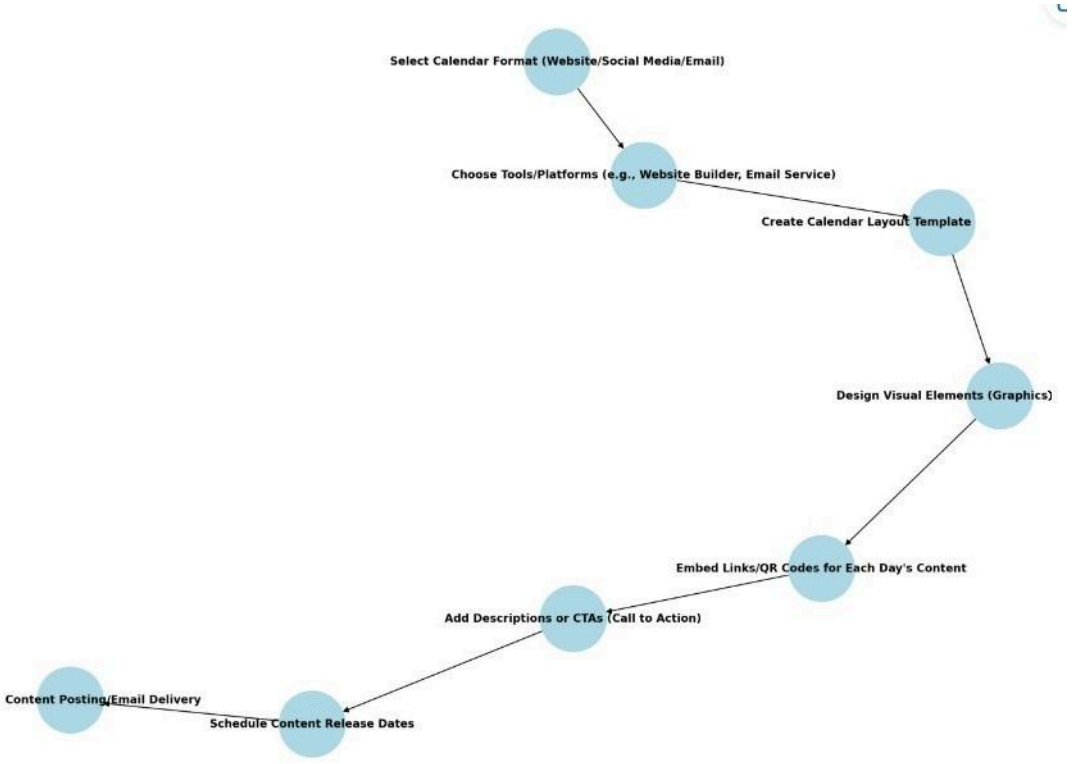


Figure 4 – Flowchart for the advent calendar solution

BPMN

BPMN (*Business Process Model and Notation*) is an helpful model for mapping any process' steps using standard business symbols, like tasks, events (start/end), gateways, and flows. This makes it easier to spot any issues like missing steps or potential bottlenecks. By using BPMN, we can make sure everything flows logically and smoothly, and if something doesn't quite fit, we can adjust it before moving forward. It's a great way to fine-tune our design and ensure it's ready for the real world. We used it as a tool to break down the three best options into smaller pieces. As instance, the model for a **music treasure hunt** for musicians to find audience would be decomposed into (at least) 7 steps, and designed as such:

- ★ **Start Event**: *listener* learns about the treasure hunt (via promotion).
- ★ **Task 1**: *listener* accesses the entry point (scan QR code, visit website).
- ★ **Task 2**: *listener* follows clues and navigates to checkpoints.
- ★ **Exclusive gateway!** ***Did the user solve the clue?***
    - ☐ Yes: Proceed to the next checkpoint.
    - ☐ No: Offer a hint or alternative path.
- ★ **Task 3**: *listener* engages with **music content** at each checkpoint.
- ★ **Task 4**: *listener* collects points or rewards at each stage.
- ★ **Task 5**: *listener* completes the final stage and unlocks the treasure.
- ★ **End event**: *listener* completes the hunt and optionally provides feedback.

Customer Journey Map

Customer journey maps are widely used to capture the complete experience of a customer interacting with a product or service. Once the single steps of a solution process (*touchpoints*) were identified, they were visualised from the user's perspective. Being a user-centric approach, the focus is on the customers' experience, emotions, and behaviors throughout their journey. This approach highlights potential challenges or opportunities for improvement, ensuring that process solutions align closely with real user experiences. CJMs are essential for refining processes, addressing pain points, and designing solutions that genuinely cater to customer needs.

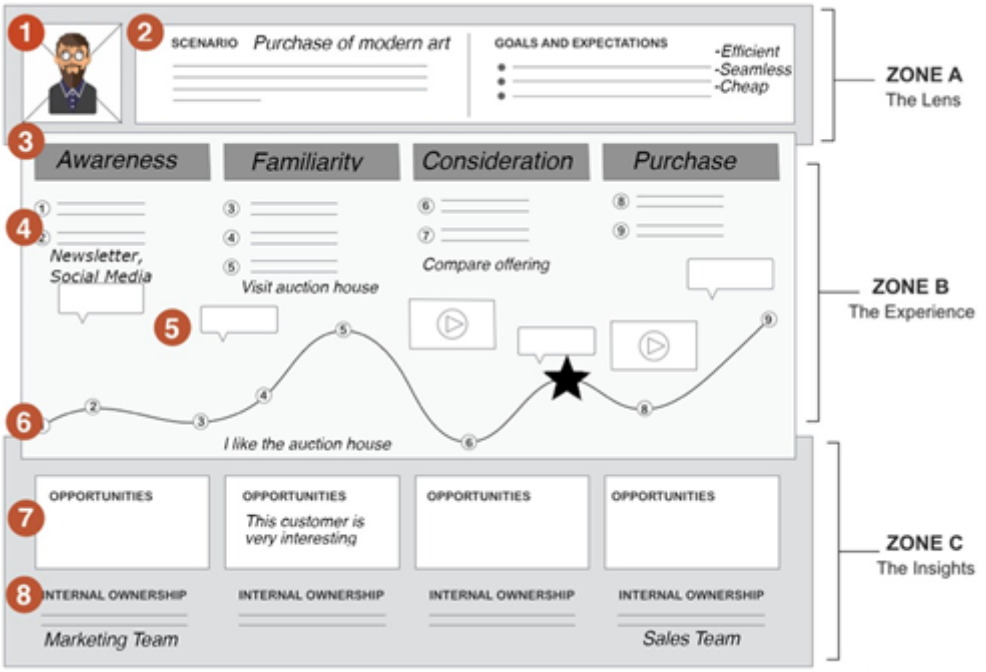


Figure 6 - Customer journey map structure

Concert postcards start by surprising customers with visually appealing postcards featuring QR codes that link to exclusive online or physical acoustic performances. The goal is to create intrigue and provide an intimate, high-quality listening experience. Once users scan the QR codes, they can enjoy the online gigs seamlessly, within existing platforms that support this kind of interaction. To keep engagement alive, follow-up actions like supporting the artist, sharing feedback, or receiving invitations to live events are essential. By focusing on simplicity, content quality, and meaningful follow-ups, concert postcards can build strong, lasting connections between emerging musicians and their audience.

Fogg Behaviour Model (B=MAP)

The *Fogg Behaviour Model* is designed to understand how user behaviour can be influenced by three key elements: following the acronym (B=MAP), Motivation, Ability, and Prompts. These factors interact in a way that determines whether a user will engage with a given solution.

1. **Motivation**: how much the user wants to take action.
2. **Ability**: how easy it is for the user to perform the desired action.
3. **Prompts**: triggers that encourage the user to take the action.

We implemented the method to our solutions to discover whether these elements were in alignment. If so, users are more likely to engage successfully with the solution. If motivation is high but the user's ability is low, simplifying the process or providing better support may help increase success. Conversely, if motivation is low, increasing the incentive or making the process more enjoyable can improve engagement.

After evaluating the **treasure hunt** option, the solution presented clear challenges in terms of feasibility. *Low user ability* was a major concern; overly complex instructions or navigation could frustrate participants and lead to dropouts. Additionally, *limited motivation* posed a risk, as users

might lose interest if rewards weren't engaging or if the experience lacked variety. Considering the significant resources required to address these issues, the treasure hunt option was deemed impractical and was discarded in favor of the remaining two solutions.

4.3.1 Results

Solution name	Description	Reason for selection	Implementation plan
Advent calendar for emerging musicians	A digital advent calendar where fans unlock exclusive songs, videos, or stories from musicians each day during a set period (e.g., 12 days).	Creates a sense of anticipation and daily engagement with minimal cost by using simple digital tools like social media or a website.	Create a calendar template, collect content from participating musicians, schedule daily posts/releases, and promote through social media or mailing lists.
Concert postcards	Musicians distribute postcards with QR codes linking to exclusive live performances.	Builds personal connections with fans through a tangible, creative approach, while being low-cost and simple to execute.	Record acoustic snippets, design postcards with free tools, print them affordably, distribute them locally, and promote fan sharing on social media.

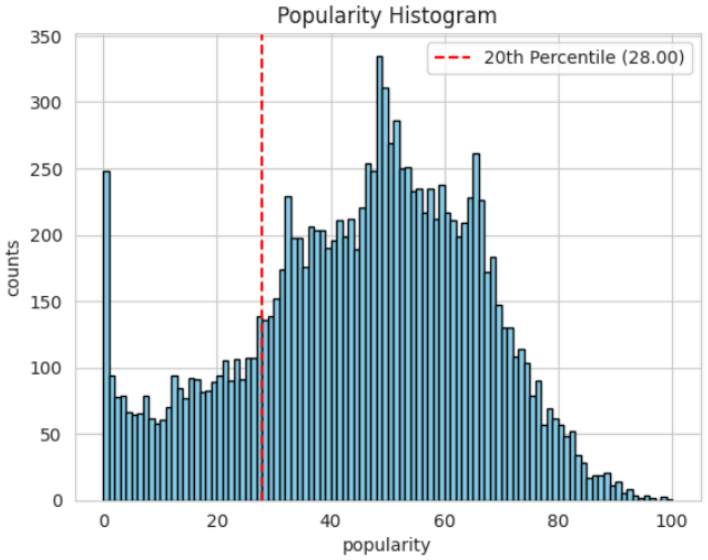


Figure 6 - popularity histogram for the artists distribution on Spotify

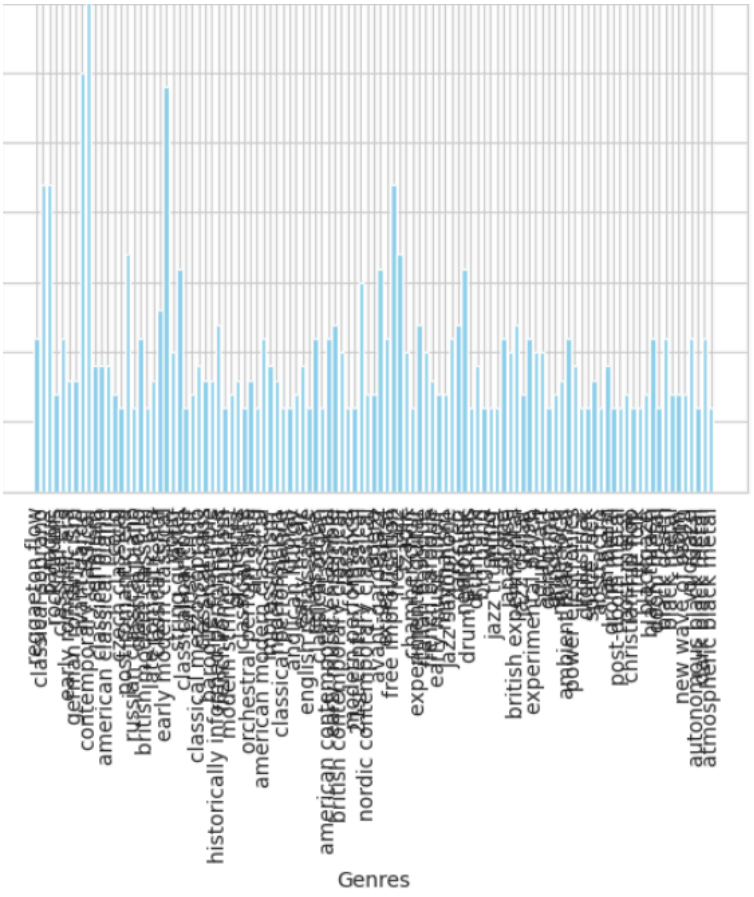


Figure 7 - Genres-popularity histogram for low-popular Spotify artists

4.4 Experimentation (Pretotyping)

The *experimentation phase* is essential for testing the feasibility of our two remaining solutions, keeping in mind the core goal: *helping emerging musicians find an audience*. During this phase, we focus on quickly validating the key assumptions behind our proposed solutions—***concert postcards*** and the ***advent calendar for emerging musicians***. By using pretotyping, we can create simple, low-fidelity versions of each solution that allow us to test whether they effectively engage the target audience. These pretotypes enable us to gather real feedback from users without significant investment, ensuring that our solutions truly resonate with emerging musicians and their potential fans. This process helps us identify areas for improvement early, ensuring that we invest time and resources in the right direction before fully getting into the *viability*.

Pretotyping

In our project, we applied pretotyping to test the technical concepts of our two solutions before committing significant resources. By developing basic versions of each solution, we could quickly gauge user reactions and validate the feasibility of the implementation. Through pretotypes, we were able to gather instant early feedback, to identify potential issues, and to make necessary adjustments before fully developing the solutions when the concepts meet user needs. For **concert postcards**, we created simple paper flyers with QR codes linking to sample music content from emerging musicians and platform profiles, allowing us to test user engagement and interest.

Similarly, for the **advent calendar**, we began by designing a minimal paper model following a template, where users could find a few featured musicians. However, implementing the digital

version revealed significant challenges due to limited time and resources. While the idea initially seemed promising, the solution required extensive technical and creative input to ensure high audience engagement. Designing visuals, interactive elements, and scheduling tools demanded intricate planning, aligned with musicians' content and permissions. Additionally, testing usability required more time than anticipated. Given these constraints, we decided to discard the advent calendar and focus on the concert postcards.

A/B Testing

A/B testing is a powerful tool we used during the pretotyping to compare two versions of our solutions and see which one works better. We were able to perform the A/B testing as soon as digital pretotypes were created. The comparison was made between the *physical models (A)* and the *digital correspondents (B)*. The team showed versions A and B to separate user groups, and was able to track the shifts in interest and gather valuable feedback. Using real user data helped in determining which option best met their needs.

4. 5 Solution Description

The *solution description* phase consolidates findings insights from earlier stages into a clear and actionable plan for implementing the chosen, most feasible solution. Following careful evaluation and testing, **concert postcards** emerged as the most practical and impactful option to connect emerging musicians with their audiences. In the following section, there are methods to outline the solution's functionalities, workflows, and required resources, serving as both a development blueprint and a communication tool to ensure smooth deployment and meaningful engagement.

Ishikawa 5M Analysis

The *Ishikawa 5M method*[9] provides a structured framework to analyze the core components of the **concert postcards** solution, ensuring that all critical factors are considered for its effective implementation.



[9] Diagramma di Ishikawa, diagramma causa effetto, le 5M e i 5Whys

Figure 5: 5M Ishikawa diagram

Man (*users*)

- *Musicians*: record acoustic performances or play online gigs; and participate in the creative process for postcard design.
- *Listeners*: engage with the postcards, scan QR codes, and provide feedback or support.
- *Designers and promoters*: ensure postcards are visually compelling and distributed effectively.

Materials (*inputs, resources*)

- Postcard paper and printing: high-quality materials for durability and aesthetic appeal.
- QR-codes: easy-to-scan links directing users to digital performances.
- Music content: high-quality acoustic recordings suitable for the target audience.

Machines (*technology, tools*)

- Design softwares: tools like Canva or Adobe Illustrator for postcard layout.
- QR-code generators: platforms for creating functional, reliable QR-codes.
- Hosting services: websites or platforms for storing and streaming and online music content.

Methods (*processes and steps*)

1. *Conceptualization*: define the design and content of postcards.
2. *Content creation*: record and edit acoustic performances.
3. *Design*: create postcard layouts incorporating branding and QR-codes.
4. *Production*: print postcards using cost-effective yet quality-assured methods.
5. *Distribution*: deliver postcards to target locations and promote online sharing.

Measurement (*success metrics*)

- **Engagement rates**: number of QR-code scans and fan feedback.
- **Reach**: distribution effectiveness and social media sharing.
- **Support metrics**: Listeners' responses, such as donations, subscriptions, or event attendance.

Environment (*context of usage*)

- **When**: postcards can be distributed at key events, like album launches or community concerts.
- **Where**: local venues, cafes, and digital campaigns to maximize reach.

5. Assessment

Assessment is crucial for evaluating the proposed solutions to ensure their effectiveness and feasibility before full-scale implementation. During this phase, the team went from measuring the suitability of the prioritized solution—**concert postcards**—against predefined criteria. These criteria typically include technical feasibility, financial viability, alignment with user needs, and the potential for long-term success. The assessment process ensures that the solution not only meets the

ultimate goals of *connecting emerging musicians with their audience* but also demonstrates sustainability and scalability. Assessment methods provide a clear picture of whether the solution is practical, cost-effective, and capable of addressing the challenges identified earlier in the project.

Pick chart

A *pick chart*[10] was used as a testing tool to assess the feasibility of the chosen prioritized solution. Evaluating the solution with regards to potential payoff and difficulty of implementation, it was clear that concert postcards represent the winner solution.

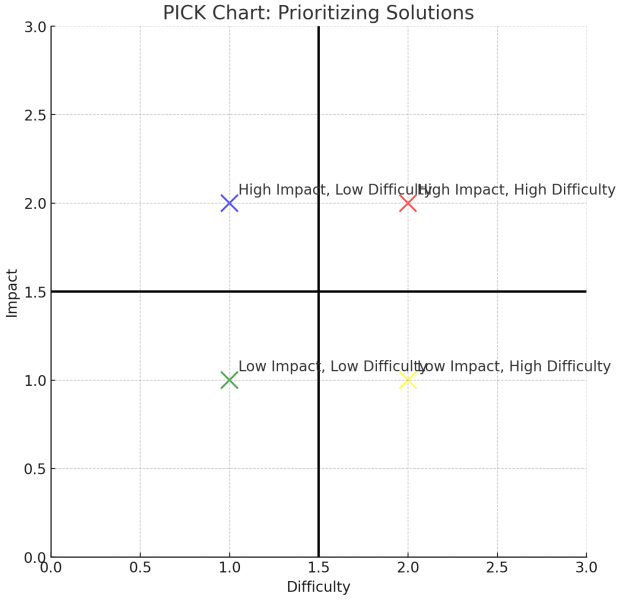


Figure 8 - Pick chart structure

Concert postcards fall into the **high payoff, low difficulty** category (blue quadrant), making them a highly favorable solution to implement. They provide a unique, engaging way for musicians to interact with their fans, leading to deeper connections and more promotion opportunities. The implementing process involves simple tasks like designing postcards, printing them, and distributing them, making it a relatively low-difficulty solution. They are easy to execute, cost-effective, and have a great potential for building connections between emerging musicians and their audiences.

NUF Test

NUF test[11] is a decision-making tool used to evaluate the solution based on three criteria:

1. **Need**: how urgent or necessary the idea is.
2. **Usefulness**: the potential value or benefit it brings.
3. **Feasibility**: how practical and achievable it is to implement.

Below is the NUF test for the **concert postcards** solution in a table format.

Criteria	Score	Analysis

[10] What is a PICK Chart: Examples and Free Template | Birdview PSA
[11] NUF Test – Gamestorming

Novelty (N)	Moderate	The concept of using postcards with QR codes linking to exclusive online or physical content is not entirely new, but the combination of physical and digital elements makes it unique. It provides a fresh way for emerging musicians to connect with their audience.
Usefulness (U)	High	The postcards are highly useful for both musicians and fans. Musicians can engage their audience with minimal cost and effort, while fans receive exclusive, engaging content that strengthens their connection with the artist.
Feasibility (F)	High	The solution is highly feasible. It requires low resources for production, including simple design tools, printing, and distribution. It is easy to implement and doesn't require significant technical expertise or financial investment.

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